Hierarchical Regionalization for Connected Redistricting Construction

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Abstract

This paper presents T.16, a hierarchical regionalization baseline for BISECT. The paper focuses on deterministic agglomeration, merge witnesses, capacity and contiguity checks, and RPLAN-compatible lineage. Regionalization is framed as a construction family that makes the sequence of region merges inspectable, not as proof that the merge sequence is optimal, legally sufficient, or politically fair.

1 Introduction

Regionalization methods construct plans by merging small connected units into larger connected regions. T.16 adds this family to BISECT so the merge sequence itself becomes part of the explanation and audit trail.

The main difference from flat capacity clustering is that regionalization has a history. A final district is not merely a set of units; it is the result of a sequence of adjacent merges. T.16 treats that sequence as evidence. The paper's central claim is therefore provenance-oriented: the constructor can emit a deterministic merge log and summary that explain how connected regions were formed under a declared capacity profile.

2 Method

The staged baseline performs deterministic agglomerative merging while tracking merge witnesses. Each witness records enough information to explain why regions were combined and to distinguish valid construction from infeasible capacity cases.

The atlas reading rule is that larger regions are not self-explanatory. The paper must show the adjacent-pair candidate set, the eligibility gate, the score used only after eligibility is established, and the merge witness that creates the next non-leaf region. Rejected adjacent pairs remain evidence when they explain why an apparently plausible merge was not taken.

2.1 Reproducible Procedure

The constructor operates on the adjacency graph emitted by the BISECT state pipeline. Connectedness is defined over that graph: candidate regions and final districts must be connected under the declared profile. Islands, disconnected components, and alternative adjacency definitions are therefore input-profile questions rather than claims solved by the merge routine.

1. Initialize each unit as a singleton region with population and adjacency metadata.
2. Enumerate adjacent candidate region pairs that remain eligible under the declared capacity and connectedness profile.
3. Score candidate merges by capacity eligibility first, then distance of merged population from the ideal district population.
4. Select the best candidate, breaking ties by merged population and stable region identifiers.
5. Record a merge witness and repeat until the requested number of regions is reached or no eligible merge remains.

Witness field	Purpose
`step`	Orders the merge in the hierarchy
`left_region`, `right_region`	Identifies the adjacent regions selected for merge
`merged_region`	Links the new region to its descendants
`merged_population`	Shows the population created by the merge
`cut_edges_between`	Records adjacency strength between the merged regions

Table 1: Implemented merge-witness fields for replayable regionalization lineage. The summary sidecar separately records merge count, hierarchy depth, capacity status, repair status, edge cut, and parameter hash.

2.2 Worked Merge Decision

The merge step is a local election among adjacent region pairs. Suppose the ideal district population is 100 with an upper capacity of 110. The constructor first separates over-capacity pairs from eligible pairs, then ranks eligible pairs by closeness to 100. A geographically adjacent pair can still lose if it would exceed capacity.

Pair	Merged pop.	Over capacity?	Distance to ideal	Decision
A+B	118	yes	18	defer: over capacity
B+C	96	no	4	selected
C+D	88	no	12	eligible, lower rank

Table 2: T.16 chooses among adjacent candidate merges. Capacity eligibility is checked before the score is used.

3 Implementation

The implementation lives in the regionalization module of `bisect-clustering` and is exposed through the regionalization structure mode. Outputs include regionalization summaries, merge logs, parameter hashes, and algorithm lineage.

The summary records the regionalization method, adjacent balanced agglomerative merge policy, repair method, capacity status, repair status, population deviation, edge cut, merge count, hierarchy depth, and parameter hash. CLI/RPLAN export also stores the merge-log path and merge-log hash in lineage extras, so a verifier can distinguish the final plan assignment from the construction history that produced it.

4 Audit Artifacts

Regionalization audit artifacts record the final assignments together with merge witnesses, parameter hashes, and algorithm lineage in the shared RPLAN and RCTX sidecar flow. The fixed-point claim is reproducibility and declared profile verification, not proof that a merge sequence is uniquely correct or legally sufficient.

The verifier can check that the exported plan and sidecars agree with the declared profile and recorded lineage. It does not certify that the merge order is globally optimal, that communities were preserved, or that the plan satisfies legal standards beyond the explicit audit profile.

The merge log is construction evidence, not an independent legal certificate. It can answer questions such as which adjacent pair merged at step 37 and what population that merge produced. It cannot answer whether that merge was the best public-interest decision, whether an unmodeled community should have been kept intact, or whether a court would accept the resulting districts.

5 Evaluation Plan

The current evidence layer includes path, grid, two-clique, capacity, and determinism fixtures, plus a benchmark-tier RPLAN package generated from the regionalization constructor on a 100-unit synthetic path. The next evidence layer should compare regionalization against flat clustering, METIS, and flow construction on selected real-data smoke states.

Claim	Evidence now	Missing evidence	Status
Connected fixture output	L0 path/grid tests	None for fixtures	Current
Deterministic merge logs	L0 determinism tests	Broader graph sweep	Current
Capacity handling	L0 capacity fixtures	Real-data smoke	Current
Audit lineage	L1 sidecar checks	Real export transcript	Current
Scale smoke	path100 benchmark package	Real-data benchmarks	Benchmark smoke
Quality vs. baselines	None in this paper	Commands and outputs	Future

Table 3: Evidence ladder for T.16 claims. Connectedness, lineage, and empirical quality are separate evidence levels.

Constructor	Primary witness	Failure artifact	Current role
Regionalization	merge witness	no eligible merge/status	baseline
Capacity clustering	repair/status	infeasibility witness	comparison
Flow construction	flow summary	infeasibility witness	comparison
Spectral	split lineage	validation failure	comparison
METIS	partition manifest	validation result	comparison

Table 4: Failure-behavior comparison framing. Quantitative comparisons remain future work until commands and outputs are recorded.

The current evidence should be read as implementation and packaging evidence: the constructor produces deterministic merge logs, summaries, and verifier-facing packages on fixtures. It is not yet evidence that regionalization produces better compactness, fewer subdivision splits, or more persuasive community outcomes than flat clustering, METIS, spectral partitioning, or flow construction.

6 Limitations

The current implementation is a deterministic regionalization baseline. It is not a complete reproduction of SKATER, Max-p, or other full spatial regionalization methods. Merge scoring choices can affect output quality and should be evaluated empirically before broad claims.

Merge lineage improves inspectability but does not make the merge sequence normatively correct. Connectedness and capacity checks do not imply compactness, community preservation, partisan fairness, or legal compliance beyond the declared audit profile.

The current scoring rule is intentionally simple: adjacent candidate pairs are ranked by capacity eligibility and population closeness, with deterministic tie-breaking. That is useful for a baseline, but it is not a reproduction of full spatial regionalization methods such as SKATER or Max-p, and it does not include richer socio-economic or community-character objectives.

7 Conclusion

T.16 makes hierarchical regionalization a first-class construction path in BISECT and preserves the same audit contract used by other final-plan producers. The contribution is replayable merge provenance under a declared profile, with empirical district-quality claims reserved for future recorded sweeps.

The paper's finished form should let a reader inspect a merge tree, understand why each local merge won, and then see exactly where the audit claim stops. That boundary is what makes regionalization useful as a construction-family member rather than a disguised optimality claim.

References

BISECT Project. T.16 hierarchical regionalization implementation spec, 2026. BISECT internal specification.